PROJECT 2

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Andrew ID: lbordign
Date: 10 Feb, 2026

The files related to the project are located in: Project2_lbordign/src

Output of 6,7,8 displayed below:
(the outputs of 1,2,3,4,5 were too large to be displayed below)

```
What would you like to do?
1: inorder
2: preorder
3: levelorder
4: postorder
5: reverseLevelOrder
6: search for points within rectangle
7: search for nearest neighbor
8: quit
> 6
Enter a rectangle bottom left (X1,Y1) and top right (X2, Y2) as four doubles each separated by a space.
> 1357605.688 411838.5393 1358805.688 413038.5393

Searching for points within (1357605.688, 411838.5393) and (1358805.688, 413038.5393)

Examined 25 nodes during search.
Found 9 crimes.

1358205.688,412438.5393,32898,5000 FORBES AV,ROBBERY,1/25/90,140100,40.44471042,-
79.94295871
1358701.856,412316.9622,32969,5050 FORBES AV,ROBBERY,4/6/90,140100,40.44441059,-
79.94116573
1358446.935,412903.5158,33181,1045 MOREWOOD AV,RAPE,11/4/90,140100,40.44600282,-
79.94213366
1358275.087,412559.1355,33347,1085 MOREWOOD AV,RAPE,4/19/91,140100,40.44504608,-
79.9427202
1358205.688,412438.5393,33570,5000 FORBES AV,AGGRAVATED
ASSAULT,11/28/91,140100,40.44471042,-79.94295871
1358205.688,412438.5393,34074,5000 FORBES AV,ROBBERY,4/15/93,140100,40.44471042,-
79.94295871
1358205.688,412438.5393,34590,5000 FORBES AV,ROBBERY,9/13/94,140100,40.44471042,-
79.94295871
1358349.449,412795.1595,33641,1060 MOREWOOD AV,RAPE,2/7/92,140100,40.44569883,-
79.94247415
1358646.638,412330.7924,35042,5044 FORBES AV,ROBBERY,12/9/95,140100,40.44444479,-
79.94136529

The crime data has been written to PGHCrimes.KML. It is viewable in Google Earth Pro.

What would you like to do?
1: inorder
2: preorder
```

3: levelorder
4: postorder
5: reverseLevelOrder
6: search for points within rectangle
7: search for nearest neighbor
8: quit
> 7
Enter a point to find nearest crime. Separate with a space.
> 1359951.000 410726.000

Looked at 27 nodes in tree. Found the nearest crime at:
1359951.481,410726.1273,32874,320 SCHENLEY RD,ROBBERY,1/1/90,140100,40.44013011,-79.93653583

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Thank you for exploring Pittsburgh crimes in the 1990's.

Screenshots

```
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Enter a rectangle bottom left (X1,Y1) and top right (X2, Y2) as four doubles each separated by a space.
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Screenshot of Google Earth taken after a rectangle search → rendering of PGHCrimes.kml.